

2023 Bi H2 Q12

Section: Sustainability and Interdependence

Topic: Components of Biodiversity

Question Summary

The question explores the components of biodiversity and the impact of habitat fragmentation caused by human activity, such as motorway construction. It asks about: (a) the two components of species diversity, (b) the bottleneck effect, and (c) the use of tunnels (corridors) to restore biodiversity.

Worked Solution

(a) Two components of species diversity

1. Species richness — the number of different species present in a habitat.
2. Relative abundance — the proportion of each species within the total number of individuals.

(Both are needed to describe the true diversity of a community.)

(b) Bottleneck effect and local extinction

- The bottleneck effect occurs when a population size is drastically reduced by events such as habitat destruction or disease.
- Only a small number of individuals survive, leading to a loss of genetic variation.
- Reduced variation limits the population's ability to adapt to

environmental change, making it more vulnerable to disease, competition, or climate shifts.

- Over time, this can lead to the local extinction of certain species.

(c) (i) Term for tunnels

Wildlife corridors (or habitat corridors).

(c) (ii) How tunnels increase biodiversity

- Corridors allow the movement of organisms between fragmented habitats.
- This increases gene flow between populations, reducing inbreeding.
- Greater gene flow and migration help restore genetic diversity and stabilise populations, supporting overall biodiversity.

Final Answer

- (a) Species richness and relative abundance
- (b) Loss of genetic variation reduces adaptability, leading to local extinction
- (c)(i) Wildlife corridors
- (c)(ii) Corridors increase gene flow, restoring biodiversity

Revision Tips

- Biodiversity includes:
 - Genetic diversity – variation within species.
 - Species diversity – variety of species and their relative abundances.

- Ecosystem diversity – range of different ecosystems.
- Habitat fragmentation → smaller, isolated populations → lower gene flow.
- Wildlife corridors reconnect habitats and maintain healthy gene pools.
- Bottleneck effect = one of the key mechanisms behind genetic drift.

Habitat A

The diagram consists of four horizontal rectangular sections stacked vertically. The top section is labeled 'Habitat A', the second 'Cleared area', the third 'Corridor', and the bottom 'Habitat B'. Each section is outlined with a thick black border.

Cleared area

Corridor

Habitat B